

Single-Agent Optimal Feedback Control for Concert-Venue Egress

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1 Introduction and scope

Given a music venue floor plan, we wish to find optimal paths and travel times between arbitrary locations on the floor to significant points of interest, mainly the stages and exits. Being able to extract these data is crucial in effective venue floor-planning, especially with attendee safety in mind. This data could, for example, provide insight into how to plan a venue such that any one guest is able to be evacuated in under some predetermined time in the case of an emergency.

Our original proposal was to model a concert crowd as a Mean Field Game (MFG), coupling a Hamilton–Jacobi–Bellman (HJB) equation for an agent’s value function to a Fokker–Planck equation for the crowd density $m(x, t)$. During our design process, however, we realized that the travel time of a single attendee from any point to another is small enough that the evolution of the crowd during this time is negligible; it is unlikely that a festival crowd redistributes itself substantially in the time it takes for an attendee to travel a few hundred meters, for example. Our project therefore differs from our proposal in a few ways: firstly, the density $m(x)$ is now prescribed and time-independent, so the Fokker–Planck half of the MFG is dropped and only the HJB is solved; and secondly, the intended outcome of the project shifts from modeling a full-venue evacuation to modeling only a single-agent extraction. Thus, we have shifted the question to: *what is the fastest route and travel time for one identified attendee from some starting position to the nearest exit?* This is the question that matters in practice when medics need to extract an unconscious concert-goer from a dense crowd, for example, and is therefore more aligned with the larger importance in solving a problem like this, namely the safety of concert attendees.

2 Problem formulation

The agent’s state $x_t \in \Omega \subset \mathbb{R}^2$ evolves under the deterministic ODE

$$\dot{x}_t = \alpha_t, \quad x_0 \in \Omega \text{ given,}$$

controlled by a velocity $\alpha_t \in \mathbb{R}^2$. The agent stops at the “hitting time”

$$\tau := \inf\{t \geq 0 : x_t \in \Gamma_e\},$$

where $\Gamma_e \subset \partial\Omega$ is the union of tagged exit segments. The cost function is

$$J(\alpha; x_0) = \int_0^\tau \left(\frac{1}{2} |\alpha_t|^2 + F(x_t, m(x_t)) \right) dt, \quad F(x, m) = \kappa \log(m(x) + \varepsilon),$$

with $\kappa, \varepsilon > 0$. These two terms encode for two physical requirements: the first is a quadratic penalty discouraging unbounded velocities, and the second is a density-aware running cost that is large in crowded regions and small in sparse ones. The value function $v(t, x) := \inf_\alpha \mathbb{E}[J \mid x_t = x]$ satisfies the (regularized) HJB

$$-\partial_t v - \frac{\sigma_{\text{att}}^2}{2} \Delta v + \frac{1}{2} |\nabla v|^2 = \kappa \log(m(x) + \varepsilon) = F(x), \quad v(T, \cdot) = 0, \quad v|_{\Gamma_e} = 0,$$

with a no-flux Neumann condition on the walls $\Gamma_w := \partial\Omega \setminus \Gamma_e$. Once v is known, the optimal velocity is recovered as follows:

$$\alpha^*(t, x) = -\nabla v(t, x).$$

3 Domain, mesh, and density field

The venue we chose to model is based on the Coachella festival grounds in Indio, CA. Our mesh is bounded over a polygonal domain $\Omega \subset (0,1)^2$ with six tagged exit segments named **Coachella Stage**, **Outdoor Theatre**, **Yuma**, **Mojave**, **Gobi**, and **Sahara**. The boundary is meshed by Gmsh with target characteristic length $h = 0.04$ using the Frontal-Delaunay algorithm; physical groups "wall" (no-flux) and "exit" (Dirichlet) are written into the .msh file and read by `GridapGmsh.GmshDiscreteModel`, producing an unstructured triangulation with 1170 triangles (Figure 1).

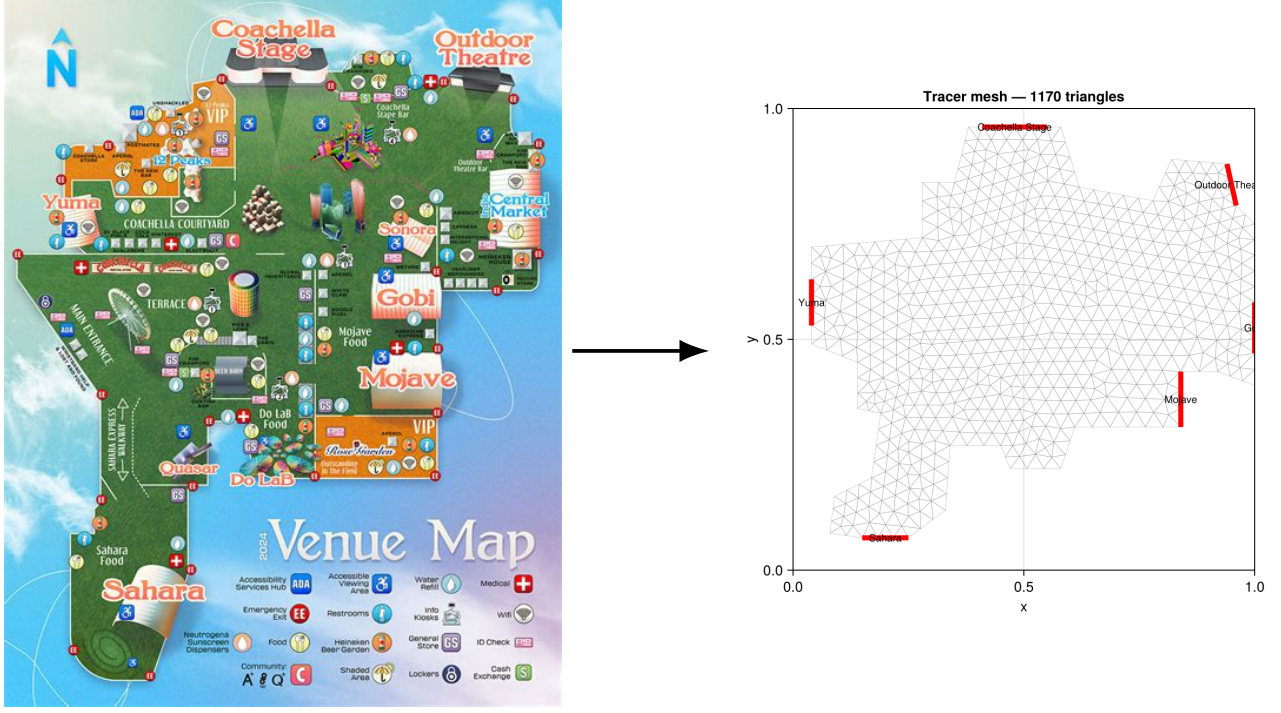


Figure 1: Left: an aerial map of the Coachella festival grounds. Right: the Coachella-like mesh used in the solver, with six tagged stage exits (red segments) where the Dirichlet condition $v = 0$ is imposed. The boundary polygon and exit locations are abstracted from the real layout, given that the exits are mainly situated by the stages.

The density field is constructed deterministically as $m(x) = m_{\text{base}} + \sum_{k=1}^K A_k \exp(-\|x - c_k\|^2 / 2w_k^2)$, with $m_{\text{base}} > 0$ guaranteeing positivity and Gaussian bumps centered at the festival's main attractions modeling denser local crowds (with $K = 10$ Gaussian bumps). We represent m as a continuous P1 finite-element function over the mesh and use it directly as RHS data in the HJB.

4 HJB discretization

Weak form. Multiplying the regularized HJB by a test function $w \in H_0^1(\Omega; \Gamma_e)$ and applying Green's identity, integrating by parts the Laplacian gives

$$\int_{\Omega} (-\partial_t v) w + \int_{\Omega} \frac{\sigma_{\text{att}}^2}{2} \nabla v \cdot \nabla w + \int_{\Omega} \frac{1}{2} |\nabla v|^2 w = \int_{\Omega} F w.$$

The boundary integral vanishes on Γ_w by the natural no-flux condition and on Γ_e by $w = 0$. The $|\nabla v|^2$ term remains a pointwise nonlinearity.

Backward implicit Euler and Newton linearization. Partitioning $[0, T]$ uniformly with $\Delta t = T/N$ and applying backward Euler at level n (sweeping $n = N-1, \dots, 0$ from the terminal condition $v^N \equiv 0$) gives, after

multiplying through by Δt ,

$$\int_{\Omega} v^n w + \frac{\sigma_{\text{art}}^2 \Delta t}{2} \nabla v^n \cdot \nabla w + \frac{\Delta t}{2} |\nabla v^n|^2 w = \int_{\Omega} v^{n+1} w + \Delta t F w.$$

This is nonlinear in v^n . We linearize by a damped Newton iteration inside each time step. Writing $v^n = v_k + \delta v$ and expanding to first order in δv , $\frac{1}{2} |\nabla v^n|^2 \approx \nabla v_k \cdot \nabla v^n - \frac{1}{2} |\nabla v_k|^2$, which is linear in the unknown v^n and yields the weak form actually solved at each Newton step:

$$\int_{\Omega} v^n w + \Delta t \frac{\sigma_{\text{art}}^2}{2} \nabla v^n \cdot \nabla w + \Delta t (\nabla v_k \cdot \nabla v^n) w = \int_{\Omega} v^{n+1} w + \Delta t F w + \frac{\Delta t}{2} |\nabla v_k|^2 w.$$

The Newton increment is scaled by a damping factor $\theta \in (0, 1]$ before updating v_k ; iteration terminates at $\|v_{k+1} - v_k\|_2 < 10^{-7}$ or at 30 iterations.

Trajectory integration. Once $v(t, x)$ is computed on the time grid, the agent’s path is obtained by forward-Euler integration of the closed-loop ODE $x_{k+1} = x_k - \nabla v(t_k, x_k) \Delta t$. The gradient is evaluated cell-wise via Gridap’s $(\nabla(v_h))(\text{Point}(\mathbf{x}, \mathbf{y}))$. Exit-crossing is detected via 2D segment–segment intersection between the integrator segment $[x_k, x_{k+1}]$ and each tagged exit segment $[a, b]$, found using Cramer’s rule. When a hit is found at parameter $t^* \in [0, 1]$ we record the travel time $\tau = t_k + t^* \Delta t$ and the exit name. This generalizes the proposal’s single right-wall exit to the multi-stage polygonal arrangement that the Coachella-like geometry requires.

5 Results

We computed the value function $v(t, x)$ once on the default density field (one HJB solve, 40 time steps), then forward-integrated trajectories from eight starting positions. Because v is the same field for every agent, this single solve answers “which exit is fastest, and how long does it take” for any starting position, thus the eight tracers in Figure 3 are merely a visualization of the algorithm’s performance on a few sample starting points.

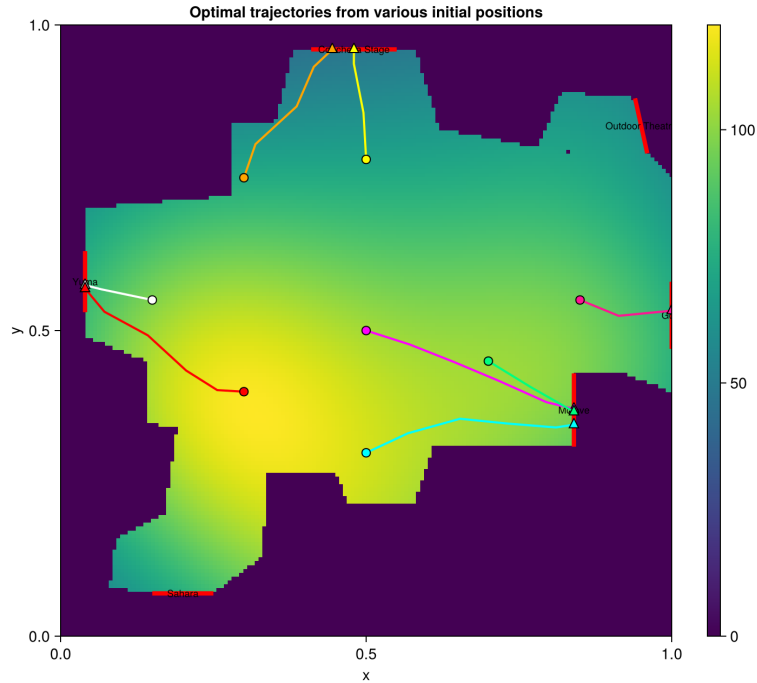


Figure 2: Eight tracers (circles = starts, triangles = exit points) following the optimal feedback $\dot{x}_t = -\nabla v(t, x_t)$ on a single HJB solve. The heatmap shows the prescribed density $m(x)$. Each tracer is routed to its nearest-in-cost exit, which is not always the nearest in Euclidean distance: e.g. the cyan tracer near $(0.5, 0.3)$ rounds the central density plateau before exiting at Mojave rather than pushing north through the densest band.

The behavior in Figure 3 matches expectation: tracers in the high-density area found towards the bottom-left of the mesh exit outward along low-density corridors, while tracers already in sparse regions exit nearly in straight lines. No tracer crosses the densest areas of the mesh unnecessarily, which is representative of the cost trade-off encoded in F .

Validation in the uniform-density limit. As a sanity check, we additionally ran validation on a uniform-density version of the same mesh as above. As expected, running this process on a uniform-density mesh yielded the straight-line path from the starting point to its nearest exit.

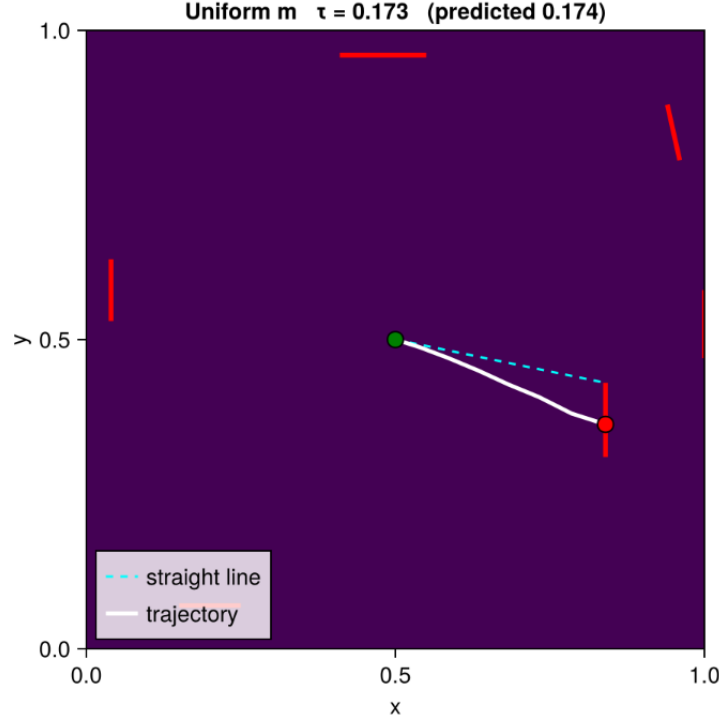


Figure 3: Validation by computing a tracer on uniform crowd density. A near-straight line is taken, as expected.

6 Discussion

Sensitivity to artificial viscosity. The stabilizer σ_{art} controls a trade-off: too small and the system diverges; too large and v is smoothed, biasing the cost-to-go upward and straightening trajectories. Empirically on $h = 0.04$, we found setting $\sigma_{\text{art}} = 0.3$ to be robust given the irregularity of the polygonal mesh.

What we gave up, what we kept, and where this is useful. In comparison to our original proposal, there is no longer a relationship between an agent and the crowd’s potential response to the decision of one agent. If this formulation were to use a true MFG, an exit would get congested precisely because agents choose to use it, and the density would adjust until a fixed point is reached. In our formulation, the density is fixed and the agent reads it as terrain, which is better fitted for the task of extracting one attendee from a crowd. What we kept is the HJB, which does almost all of the algorithmic work in both formulations. The single-agent model already answers a concrete operations question: given an attendee’s reported location and a snapshot of crowd density, one HJB solve produces an optimal feedback field $-\nabla v$ over the entire venue, along which a medic can be sent with an estimated travel time τ that accounts for the crowd in between.

Possible future extensions Further extensions include reincorporating the Fokker-Planck formulation to model the crowd density as an MFG that slowly evolves over time. This would allow us to model both individual movements

as well as large-scale crowd shifts, which result from things like emergencies and entrances/exits at the beginning and end of events. This would provide further insight into things like venue design and operations management.

7 Code

All code used for this project, including the simulation scripts, plotting utilities, and implementation details, is available in our [GitHub repository](https://github.com/Yah-Boi-1/Mean-Field-Crowd-Dynamics): <https://github.com/Yah-Boi-1/Mean-Field-Crowd-Dynamics>.

References

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